Spatial–Spectral–Temporal Deep Regression Model With Convolutional Long Short-Term Memory and Transformer for the Large-Area Mapping of Mangrove Canopy Height by Using Sentinel-1 and Sentinel-2 Data

Ilham Jamaluddin, Ying-Nong Chen, and Kuo-Chin Fan

***Abstract*— Mangrove canopy height information is crucial to carbon stock and biomass analyses. However, estimation of this height is challenging because of the large areas involved, and the field conditions of mangrove forests. Remote sensing satellite imagery has been used for canopy height mapping because it offers several advantages. This study developed a spatial–spectral–temporal deep learning regression model with convolutional long short-term memory (ConvLSTM) and transformer (hereafter referred to as the SST-CLT model) to map mangrove canopy height over a large area. The SST-CLT model consists of two submodels trained simultaneously. The first submodel is a fusion extractor to extract spatial–spectral–temporal information from Sentinel-1 time-series data by using a ConvLSTM. It also extracts spatial–spectral information from Sentinel-2 data using a 2-D convolutional block. The second submodel is a regressor that contains a Swin transformer (SWINTF) and a final convolutional regression layer. Data from the light detection and ranging (LiDAR) canopy height model (CHM) were employed as the target data to train the proposed model. The SST-CLT model was tested on two datasets collected from Florida: a large dataset for the Everglades National Park (ENP) and a small dataset for the Charlotte Harbor Preserve State Park (CHPSP). The SST-CLT model achieved a mean absolute error (MAE) of 1.924 and 1.913 m for the ENP and CHPSP datasets, respectively. Moreover, it achieved root mean square error (RMSE) values of 2.471 and 2.440 m for these datasets, respectively. The SST-CLT model was compared with that of other regression models. The results indicated that the MAE and RMSE of the proposed SST-CLT were lower than those of the other models (https://github.com/ilhamjamaluddin/SST-CLT).**

***Index Terms*— Canopy height, deep learning, light detection and ranging (LiDAR), mangrove, regression, Sentinel-1, Sentinel-2.**

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I. Introduction

MANGROVE forests are located in tropical and subtropical coastal tidal zones [1]. These forests provide several benefits to natural ecosystems, such as coastal protection and storage of carbon in vegetation [2]. Moreover, they are considered a blue carbon ecosystem, which has a high carbon stock and helps reduce CO_2 emissions in the atmosphere [3]. Mangrove forests are also recognized as the most carbon-rich type of forest in tropical areas [4]. Above-ground biomass (AGB) is considered the most useful quantitative measure of the carbon content of mangroves [5], [6]. Knowledge of tree canopy height is essential for estimating the amount of biomass in a tropical forest [7], and this height is correlated with critical forest biophysical parameters, including AGB [8].

Estimating mangrove canopy height is challenging because of the large area and limited accessibility of mangrove forests. Direct measurements of canopy height in large areas—for example, in situ observations as well as airborne and space light detection and ranging (LiDAR)—are inefficient and costly [9]. Airborne or space LiDAR is considered the most straightforward technology for measuring canopy height in large areas [10]. Airborne LiDAR or airborne laser-scanning data can be used in forest measurements, such as canopy height estimation [11]. The use of the Ice, Cloud, and land Elevation Satellite (ICESat) in mapping global forest canopy height is an example of using space LiDAR [12]. However, the use of LiDAR data in forest canopy height estimation has some limitations, such as limited area coverage, high cost, and limited temporal coverage [13], [14], [15]. To overcome the aforementioned limitations, remote sensing satellite imagery, which provides large-scale observations and is publicly available, has been used for mangrove canopy height estimation. Remote sensing images have also been widely employed for tree canopy height mapping in large areas. A common technique for estimating tree canopy height from multispectral satellite imagery is regression modeling, which involves the use of in situ data or LiDAR canopy height data as reference data [16], [17].

With advances in regression techniques, machine learning and deep learning have been used for estimating tree canopy

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height in mangrove forests and other types of forests. Machine learning regression techniques, such as regression trees and random forest (RF), are commonly used to estimate tree canopy height in a forest by integrating satellite imagery with LiDAR canopy height data or in situ measurements [18], [19], [20], [21]. For example, Hansen et al. [18] employed Landsat 7, Landsat 8, and Geoscience Laser Altimeter System (GLAS) data in a regression tree algorithm for tree canopy height mapping in sub-Saharan Africa. Moreover, Wang et al. [19] used Moderate Resolution Imaging Spectroradiometer (MODIS) and GLAS data in an RF algorithm to estimate the global mean canopy height of a forest. The results of the aforementioned sets of authors have indicated that regression tree and RF algorithms can be successfully used on optical satellite imagery and GLAS data for forest canopy height estimation. Li et al. [20] employed an RF algorithm together with ICESat-2 data and optical and synthetic aperture radar (SAR) satellite imagery. Their results indicated that combining the backscattering coefficients for the SAR data and the red-edge-related variables for the optical data positively contributed to tree canopy height estimation. Ghosh et al. [21] used optical and SAR imagery with field tree height data to compare the performance of RF and symbolic regression in mangrove canopy height estimation, and they reported favorable results with symbolic regression outperforming RF in this study. Pham and Brabyn [22] conducted RF regression on SPOT images for AGB estimation.

As a subset of machine learning, deep learning has been widely employed in the analysis of remote-sensing satellite imagery [23]. Rapid developments have been achieved in convolutional neural networks (CNNs), and such networks can provide technical support for the analysis of remote-sensing satellite images. CNNs have been used in several tasks, such as image classification [24], pixel segmentation [25], object detection [26], and continuous variable regression [27]. One advantage of CNNs is that they are capable of processing multiband remote-sensing images to obtain large-scale environmental information [23]. CNNs have been widely used for the semantic segmentation of mangrove pixels. For example, Guo et al. [28] used the capsule U-Net network to conduct large-scale mangrove mapping on the Maritime Silk Road between 1990 and 2015.

Because of the locality of their convolution operation, CNNs have some drawbacks, such as an inability to directly model the interaction between global semantic and context information [29]. To solve this problem, Fu et al. [30] employed an attention mechanism. In addition, visual transformers based on the original transformer architecture used in natural language processing have been successfully employed for modeling global relationships [31], [32]. For example, Liu et al. [33] used the Swin transformer (SWINTF), which utilizes shifted windows, for modeling global relationships and achieved favorable results. In other studies, CNNs have been combined with the SWINTF for the segmentation of medical images [34] and remote sensing [35] images.

Most studies that have conducted deep learning on remote sensing satellite imagery have performed classification tasks (land use or land cover mapping), with only a few studies employing deep learning in regression tasks. Shah et al. [36] adopted CNNs for canopy height estimation by using LiDAR and Landsat data. Rodriguez and Wegner [37] proposed a new method for deep tree density estimation in which Sentinel-2 images are used. Lang et al. [9] employed separable convolution and Sentinel-2 data for country-wide (Switzerland and Gabon) vegetation height mapping. Finally, Wang et al. [38] employed deep CNNs and dual-polarized Radarsat-2 satellite images for estimating sea ice concentration.

Considering the aforementioned problems, this study developed a new spatial–spectral–temporal deep regression model with a convolutional long short-term memory (ConvLSTM) and transformer (hereafter referred to as the SST-CLT model) for estimating mangrove canopy height from Sentinel-1 (SAR) time-series data and Sentinel-2 (Optical) data. Studies have obtained favorable results in canopy height estimation when using optical and SAR imagery [20], [21], [39]. Data from the LiDAR canopy height model (CHM) developed by the National Aeronautics and Space Administration (NASA) were used as the target data to train the model proposed in this study.

The SST-CLT model generally consists of two submodels that are trained simultaneously. The first submodel is a spatial–spectral–temporal fusion extractor and the second submodel contains the SWINTF and a convolutional regression layer. The SST-CLT model can perform large-scale estimation of mangrove canopy height at a spatial resolution of 10 m. This model was tested using two datasets collected from FL, USA: a large dataset collected for the Everglades National Park (ENP) and a small dataset collected for the Charlotte Harbor Preserve State Park (CHPSP).

The main contributions of our work are as follows.

1) *Developed a deep learning regression model (SST-CLT):* We have effectively combined the convolutional, ConvLSTM, and SWINTF architecture for remote sensing image regression task.
2) *Fusion extractor submodel:* To effectively combine the Sentinel-1 time series and Sentinel-2 data, we constructed a fusion extractor submodel that consists of a spatial-spectral-temporal extractor part for the Sentinel-1 time series (SST-S1) and spatial-spectral extractor part for Sentinel-2 (SS-S2).
3) *Effective utilization ofSWINTF for mangrove canopy height estimation:* To extract the local mangrove pixel height value, we fed the extracted feature maps from the fusion extractor submodel to the SWINTF architecture and then used a linear pointwise 2-D convolutional as the final layer to produce a single canopy height per pixel. The ablation study result demonstrates the SWINTF effectively extracts the local mangrove pixel height information.
4) *Extensive experiments based on two datasets:* We produced two mangrove canopy height datasets for large and small datasets. The results demonstrate the superiority of the proposed model over other state-of-the-art regression algorithms in terms of mangrove canopy height error values on both datasets. We also produced

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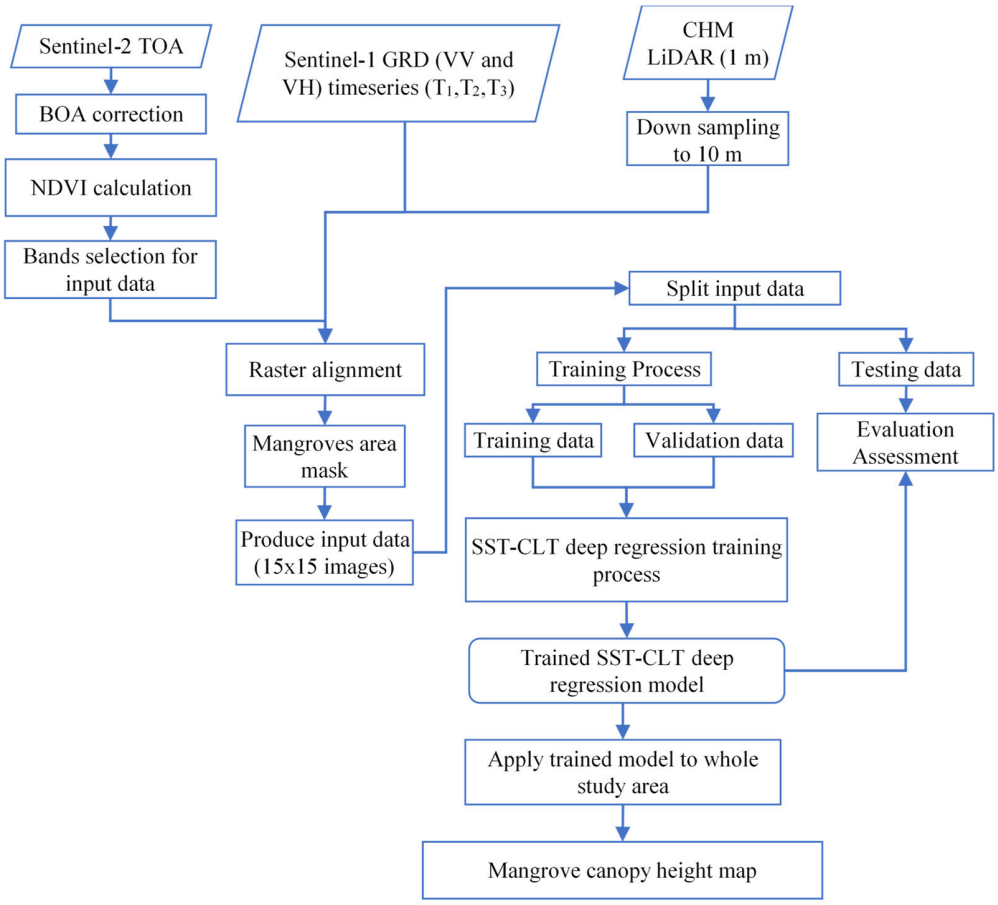


Fig. 1. Processing flowchart of this study. TOA, top of atmosphere; BOA, bottom of atmosphere; NDVI, normalized difference vegetation index; and CHM, canopy height model.

two mangrove canopy height maps in ENP and CHPSP areas in 2017.

The rest of the article is organized as follows. The materials and methods used in this study, including the study area, datasets, processing flow, proposed SST-CLT deep regression model, and model evaluation assessments, are detailed in Section II. The experiments and experimental results are described in Section III. A discussion on the results is provided in Section IV. Finally, the conclusions of this study are presented in Section V.

II. Materials and Method

The processing flow used in this study is displayed in Fig. 1. Sentinel-2, Sentinel-1, LiDAR CHM, and mangrove cover mask data were used as input data. Because all the input data had a spatial resolution of 10 m, the final predicted mangrove canopy height also had a spatial resolution of 10 m. The Sentinel-2 data contained the blue, green, red, near-infrared (NIR), short-wave infrared 1 (SWIR-1), short-wave infrared 2 (SWIR-2), and normalized difference vegetation index (NDVI) bands. In contrast, the Sentinel-1 data contained a time series of backscattering values for images captured under vertical transmission and vertical receipt (VV) and under vertical transmission and horizontal receipt (VH). The preprocessing process involved NDVI calculation, raster alignment, the production of 15 × 15-pixel images, and input data splitting. The 15 × 15-pixel images were split into training, validation, and testing data. The training and validation data were used for training the SST-CLT model, whereas the testing data were used for model evaluation. The trained SST-CLT model was then employed to predict the mangrove canopy height in the entire study area.

A. Study Areas

Two study areas were considered in this research: the ENP and CHPSP (see Fig. 2). These areas were employed to verify the performance of the SST-CLT model when using a large dataset (ENP) and a small dataset (CHPSP). The small red boxes in Fig. 2 represent 15 × 15-pixel input data along Goddard's LiDAR, Hyperspectral, and Thermal (G-LiHT) trajectories. The LiDAR CHM data obtained from the G-LiHT imagery of the two study areas were used as the target data. Details on the datasets used in this study are provided in Section II-B.

In the ENP, the total mangrove area is 120 124 ha [40]. This study used this area as a mangrove mask for the ENP. The ENP has three types of mangrove: *Laguncularia racemosa*(white mangrove), *Avicennia germinans*(black mangrove), and *Rhizophora mangle*(red mangrove) forests [5]. Based on the LiDAR CHM data obtained from G-LiHT trajectories, the

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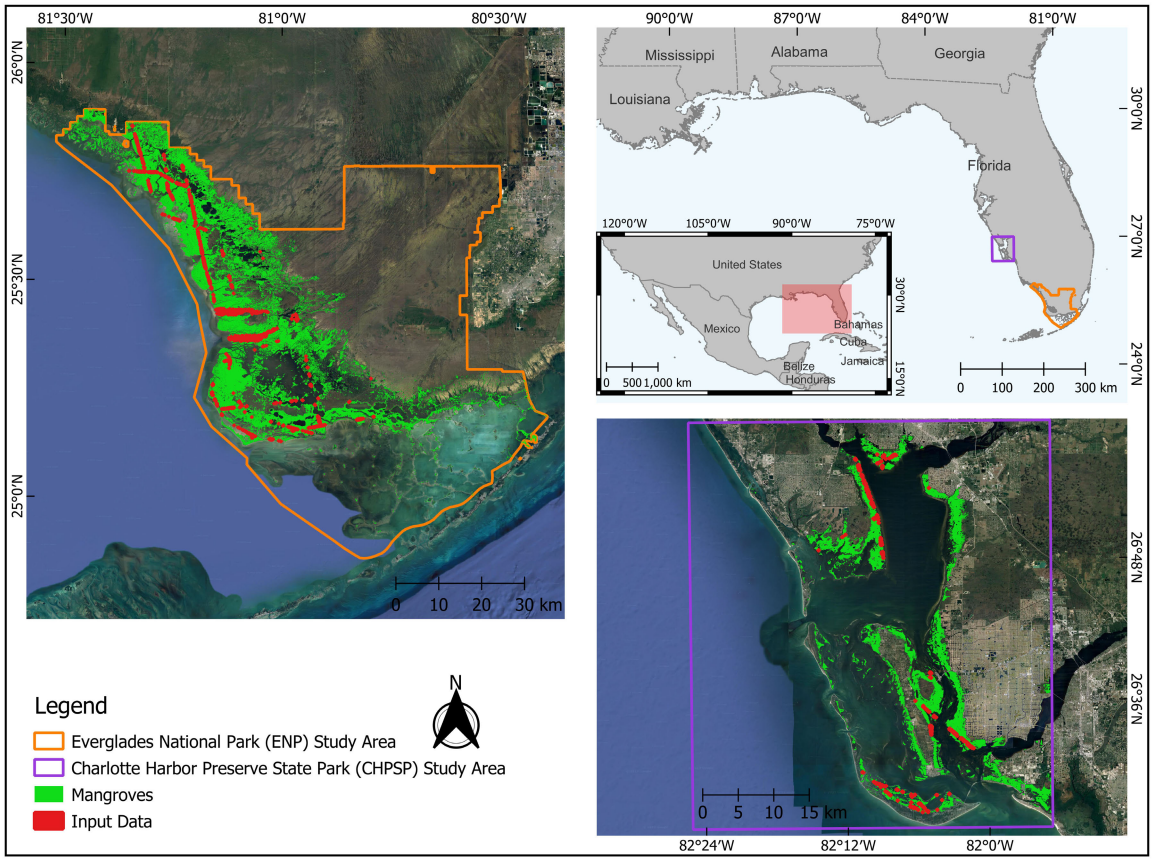


Fig. 2. Two study areas, namely the ENP and CHPSP, and distribution of the input datasets along the G-LiHT trajectories.

average, minimum, and maximum mangrove heights in the ENP were determined to be 11.27, 0.07, and 25.68 m, respectively.

According to Global Mangrove Watch (GMW), the total mangrove area in the CHPSP was 18 915 ha in 2017 [41]. A study identified five types of mangroves species in the CHPSP by performing Landsat data mapping: red mangrove, black mangrove, white mangrove, red–black–white mangrove mixes, and black–white–buttonwood mangrove mixes [42]. Red–black–white mangrove mixes are the dominant mangrove type in the aforementioned study area [42]. On the basis of the LiDAR CHM data obtained from G-LiHT trajectories, the average, minimum, and maximum mangrove heights in the CHPSP were determined to be 4.85, 0.03, and 17.87 m, respectively.

B. Datasets

1) Satellite Dataset: Sentinel-1 and Sentinel-2 data were used as the input data in the proposed model; the adopted input data are presented in Table I. For the ENP, a mosaic of four Sentinel-2 scenes was used to cover the entire area. The scene codes of the Sentinel-2 data and the corresponding acquisition dates were as follows: T17RMJ—December 30, 2016; T17RNJ—January 6, 2017; T17RMH—February 15, 2017; and T17RNH—February 15, 2017. The acquisition date selected was close to March 2017 for which the cloud cover percentage was minimal. For the CHPSP, the scene code was T17RLK, and the date on which the Sentinel-2 data were acquired was March 10, 2017.

TABLE I
SENTINEL-1 TIME-SERIES DATA AND SENTINEL-2 DATA USED IN THIS STUDY

Data	Features	Time-series	Description
Sentinel-1	σVV σVH	ENP: 02/03/2017 14/03/2017 26/03/2017 CHPSP: 07/03/2017 19/03/2017 31/03/2017	σ stands for backscattering coefficient.
Sentinel-2	Blue (B2) Green (B3) Red (B4) NIR (B8) SWIR-1 (B11) SWIR-2 (B12)	-	Additional bands (NDVI) was calculated based on Red and NIR bands

Sentinel-2 images were downloaded from the Sentinels Scientific Data Hub (https://scihub.copernicus.eu/dhus/, accessed on December 1, 2021). Level 1C Sentinel-2 images (top-of-atmosphere product with radiometric and geometric correction

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in accordance with a global reference system) were downloaded and corrected to a bottom-of-atmosphere reflectance product by using the Sen2Cor software developed by the ESA. Subsequently, the blue, green, red, NIR, SWIR-1, and SWIR-2 bands were selected. To ensure uniform spatial resolution over the entire dataset, the spatial resolution of the SWIR-1 and SWIR-2 bands was changed from 20 to 10 m. We then mosaicked all the Sentinel-2 images to cover the entire study area. In addition, the NDVI was calculated [see (1)]. The NDVI indicates the greenness level of vegetation and is widely used for vegetation mapping because of its advantages, such as can enhance the vegetation pixels [39], [40]

$$\frac{\text{NIR} - \text{Red}}{\text{NIR} + \text{Red}}. \tag{1}$$

The time-series data of Sentinel-1 from March were also used as the input data, and the C-band SAR data of Sentinel-1 were obtained from the Google Earth Engine. SAR time-series information can also help reveal the average canopy height in a study area during a certain time. The combination of SAR and optical data enhances the accuracy of canopy height estimation [20]. The level-1 ground range detected dual-polarized backscattering coefficients (σ) of VV and VH Sentinel-1 images were also used as the input bands from Sentinel-1. In summary, this study employed three time series of Sentinel-1 data for the entire month of March 2017 for the ENP and CHPSP (see Table I).

2) LiDAR CHM: The SST-CLT model is a supervised deep-learning regression architecture, and the target data of this model was LiDAR CHM data obtained from G-LiHT measurements of NASA [43]. The G-LiHT measurements for the ENP and CHPSP were made on March 22–29 and March 27, 2017, respectively. The G-LiHT level-3 products were used to obtain LiDAR CHM data. A LiDAR CHM was constructed by selecting the greatest return height in each cell of a 1-m grid, and these points were used to create a triangular irregular network and interpolate the canopy height on a 1-m raster grid [43]. Because the original spatial resolution of the LiDAR CHM was 1 m, all LiDAR CHM data were resampled to a spatial resolution of 10 m by using a bilinear algorithm so that the input data (LiDAR CHM, Sentinel-1, and Sentinel-2 data) all had the same spatial resolution.

3) Input Data Processing: A total of 12 flight trajectories around the ENP and 13 flight trajectories around the CHPSP were downloaded and used to obtain CHM data. Raster alignment was then performed to equalize the pixel positions of the Sentinel-2, Sentinel-1, and LiDAR CHM data. Subsequently, all Sentinel-1, Sentinel-2, and LiDAR CHM data for the ENP and CHPSP were masked using mangrove cover data obtained from a previous study [40] and GMW [41], respectively.

Because of the narrow trajectory of the LiDAR CHM, to maintain a constant input image size for the proposed model, all the Sentinel-1, Sentinel-2, and LiDAR CHM data were cropped to a size of 15 × 15 pixels. The cropped images had to meet two criteria (see Fig. 3): 1) they could not contain null CHM values, and 2) all their CHM values had to be higher than 0 m. The spatial distributions of the 15 × 15-pixel input images obtained for the study areas are displayed in Fig. 3.

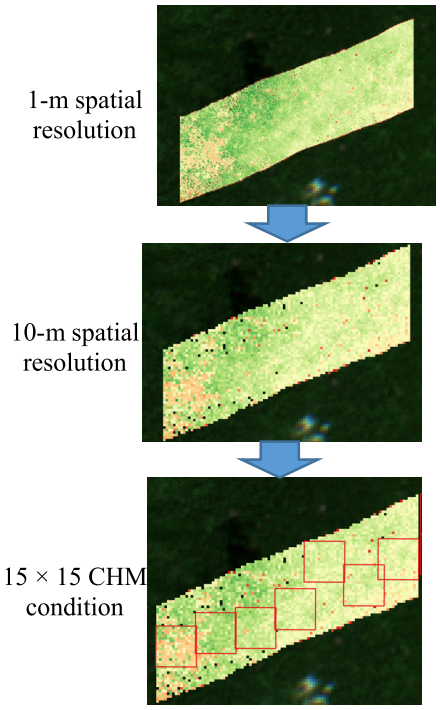


Fig. 3. Cropping of input images to a size of 15 × 15 pixels.

TABLE II
DISTRIBUTION OF THE ENP DATASET

	Total Images (15x15)	Total Pixels	Mean CHM (m)
Training	794	178,650	11.23
Validation	198	44,550	11.49
Testing	249	56,025	11.31
Total	1,241	279,225	11.27

A total of 1241 cropped images were obtained for the ENP, and these images constituted the large dataset for this research. This dataset comprised 279 225 pixels with a spatial resolution of 10 m (see Table II). For training the proposed model, 80% and 20% of the aforementioned data were separated into a training set and testing set, respectively (992 and 249 images, respectively). The testing dataset comprised novel independent data that were not used in the training process and were used to evaluate the trained model. The training data were separated into 80% for the training set (794 images) and 20% for the validation set (198 images) to achieve early stopping and prevent overfitting. The mean mangrove CHM values obtained using the training, validation, and testing datasets were almost the same, that is, approximately 11 m.

A total of 237 cropped images were obtained for the CHPSP, and these images constituted the small dataset for this research. This dataset comprised 53 325 pixels (see Table III) and was divided as described in the aforementioned text, with the training, validation, and testing sets containing 152, 37, and 48 images, respectively. The mean CHM values obtained using the training and validation data were almost the same,

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TABLE III
DISTRIBUTION OF CHPSP DATASET

	Total Images (15x15)	Total Pixels	Mean CHM (m)
Training	152	34,200	4.628
Validation	37	8,325	4.638
Testing	48	10,800	5.722
Total	237	53,325	4.996

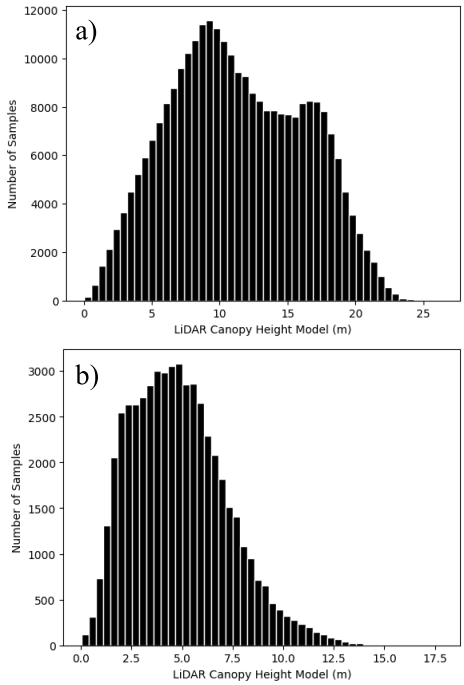


Fig. 4. Histograms of canopy height for the (a) ENP and (b) CHPSP.

approximately 4.63 m, whereas the mean CHM value obtained using the testing data was 5.72 m.

Histograms of the mangrove canopy height for the ENP and CHPSP are displayed in Fig. 4. For the ENP, the canopy height mainly exhibited a normal distribution, with few values being less than 1 m and higher than 22 m. For the CHPSP, the distribution of canopy heights was positively skewed, with most values being between 0 and 7 m and few values being greater than 7 m.

C. SST-CLT Deep Regression Model

The SST-CLT model was proposed and considered the spatial-spectral-temporal correlation of Sentinel-1 time series and Sentinel-2 data, and LiDAR CHM was used as the target data. In general, the SST-CLT has two submodel: 1) fusion extractor submodel that consists of SST-S1 extractor part and SS-2 extractor part, and 2) SWINTF regressor that exploit the advantages of SWINTF then fed to the final regression layer.

1) Fusion Extractor: The main purpose of the fusion extractor submodel is to extract spatial–spectral–temporal feature information from Sentinel-1 time-series and Sentinel-2 data. This submodel consists of two extractors: SST-S1 and SS-2.

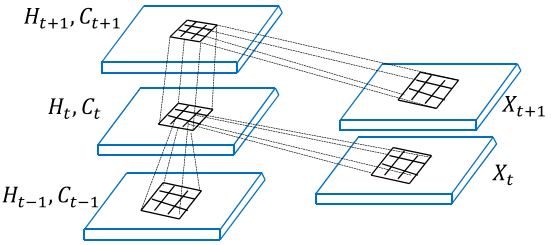


Fig. 5. Inner structure of the 2DConvLSTM network [44].

The SST-S1 extractor uses Sentinel-1 SAR time series as input data, whereas the SS-2 extractor uses Sentinel-2 multispectral images as input data.

a) SST-S1 extractor: The collected Sentinel-1 data comprised C-band signals, which can penetrate the crown of vegetation and provide useful information for canopy height estimation. Such signals also pass through clouds, thereby providing cloud-free, time-series data. The SST-S1 extractor contains a 2-D ConvLSTM (2DConvLSTM) network [44]. This network is an extended version of a long short-term memory (LSTM) network and considers a spatial relationship using convolutional structures. By contrast, an LSTM network considers only spectral–temporal correlations. Because of this advantage of the SST-S1 extractor, it can exploit the spatial–temporal–spectral correlations of Sentinel-1 time-series data to determine the average canopy heights in the study areas during a certain time period. The inner structure of the 2DConvLSTM network is expressed as follows [44]:

$$\begin{aligned}
&\text{2DConvLSTM} = \\
&i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + W_{ci} \circ C_{t-1} + b_i), \\
&f_t = \sigma\left(W_{xf} * X_t + W_{hf} * H_{t-1} + W_{cf} \circ C_{t-1} + b_f\right), \\
&C_t = f_t \circ C_{t-1} + i_t \circ \tanh\left(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c\right), \\
&o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + W_{co} \circ C_t + b_o), \\
&H_t = o_t \circ tanh(C_t)
\end{aligned} \tag{2}$$

where $X_1, \ldots, X_t$ are the inputs; $C_1, \ldots, C_t$ are the cell outputs; $H_1, \ldots, H_t$ are the hidden states; i_t, f_t, and o_t are the gates of the ConvLSTM network and are 3-D tensors; "$*$" denotes the convolutional operator; and "$\circ$" denotes the Hadamard product. The inner structure of the 2DConvLSTM network is displayed in Fig. 5.

The SST-S1 extractor consists of three 2DConvLSTM layers with a kernel size of 3×3. The input of these layers is a 5-D tensor denoted $X_n \times T \times R \times C \times$D, where X_n, T, R, C, and D are the total number of input images, time or sequence, input width, input height, and number of channels or bands, respectively. Sentinel-1 time series data were used as input data for these layers in this study. Therefore, T was 3, both R and C were 15, and D was 2 (VV and VH bands). The total number of filter maps (or dimensionality) of the output space for the three 2DConvLSTM layers was 64, 128, and 256, respectively. The feature maps obtained from the first and second 2DConvLSTM layers comprised a 5-D tensor with sequence data, whereas the output feature maps obtained from the third 2DConvLSTM layer comprised a 4-D tensor without sequence features. Therefore, the output of the SST-

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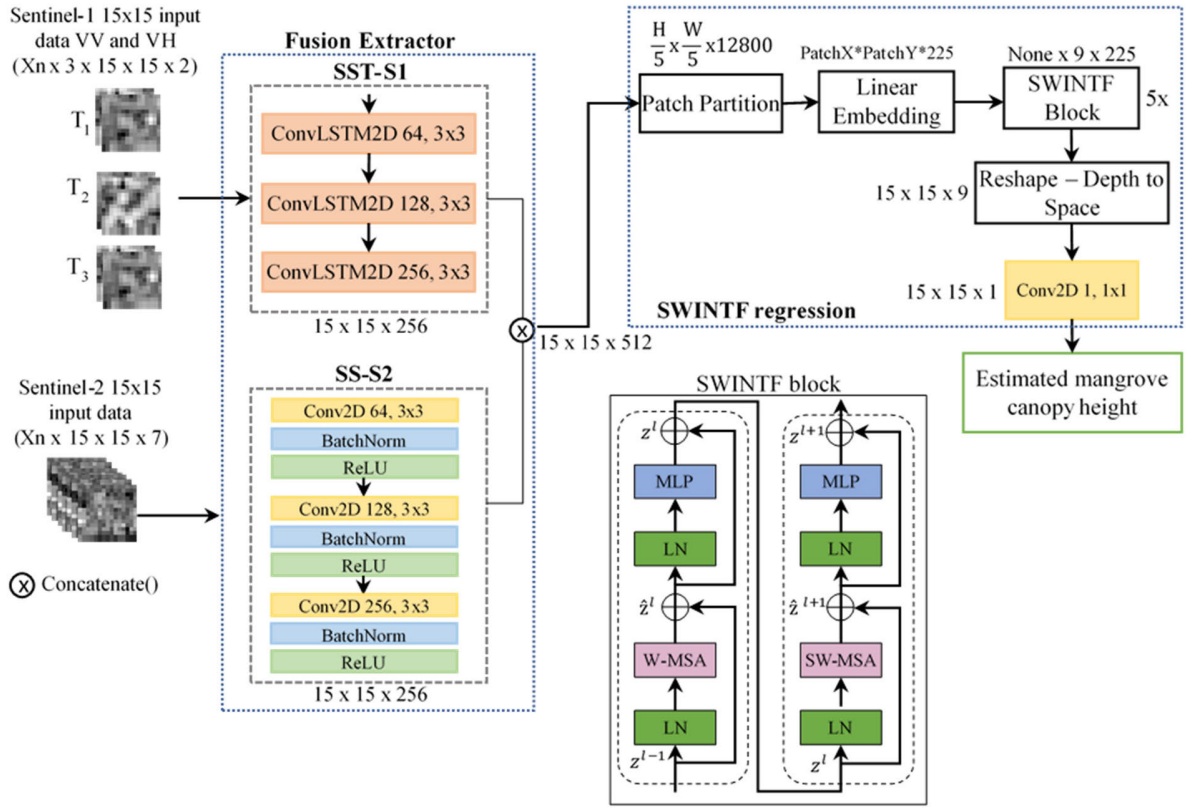


Fig. 6. Architecture of the proposed SST-CLT deep regression model.

S1 extractor could be expressed as follows:

$$X_{\text{fm,2DConvLSTM},3,64}(T, R, C, D) = \{X_n(3, 15, 15, 64)\} \quad (3)$$

$$X_{\text{fm,2DConvLSTM},3,128}(T, R, C, D) = \{X_n(3, 15, 15, 128)\} \quad (4)$$

$$X_{\text{fm,2DConvLSTM},3,256}(T, R, C, D) = \{X_n(15, 15, 256)\} \quad (5)$$

where $X_{\text{fm,ConvLSTM},n,f}$ represents the ConvLSTM feature map and n and f represent the kernel size and total number of filter maps, respectively.

b) SS-S2 extractor: Studies have confirmed that a correlation exists between optical imagery and canopy height [9], [18], [22]. Thus, the SS-S2 extractor uses Sentinel-2 optical images as input data. This extractor employs a 2-D convolutional layer to exploit the spatial–spectral information embedded in Sentinel-2 optical imagery. The output feature maps of this layer are denoted $X_{\text{fm,2DConv}}(R, C, D)$, where R, C, and D indicate the total width, height, and number of channels or bands, respectively. The SS-S2 extractor contains three typical convolutional blocks, namely a 2-D convolutional layer followed by a batch normalization layer and a rectified linear unit (ReLU) activation function (f) layer. The ReLU activation function layer is used to increase data nonlinearity; the calculation is as follows:

$$f(X_{\text{fm,2DConv}}(R, C, D) = \max\{0, X_{\text{fm,2DConv}}(R, C, D).\} \quad (6)$$

The batch normalization layer is used to normalize the data feature maps, which results in a more effective and faster learning procedure [45]. Moreover, the 2-D convolutional layer contains 3 × 3 kernels. The 2-D convolutional layer, batch normalization layer, and activation function layer use 64, 128, and 256 filter maps, respectively.

Because the goal of the fusion extractor submodel is to exploit the spatial–spectral–temporal information from Sentinel-1 time-series data and Sentinel-2 data, the feature maps output by the SST-S1 and SS-S2 extractors are concatenated as follows:

$$X_{\text{fm,cfe}} = S\{[X_{\text{fm,2DConvLSTM}}, X_{\text{fm,2DConv}}]\} \quad (7)$$

where $X_{\text{fm,cfe}}$ denotes the concatenated feature map and S represents the concatenation layer. The feature maps output by the fusion extractor submodel are used as the input for the SWINTF submodel.

2) SWINTF Submodel: The SWINTF [33] submodel uses nonoverlapping shifted windows to calculate attention locally so that detailed spectral information can be obtained to improve the performance of the proposed model. Moreover, this submodel reduces the computational complexity and allows for a cross-window connection by using a shifted-window scheme.

The SWINTF submodel comprises four main parts: a patch partition module, a linear embedding module, five SWINTF blocks, and a reshaping module. The concatenated feature maps output by the fusion extractor submodel is first input to the patch partition module. This module then splits each input feature map, which has a size of 15 × 15 pixels, into nine

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nonoverlapping patches of size 5 × 5 pixels, each of which is treated as a token. Subsequently, the output of the token by the patch partition module are input to the linear embedding part. Next, the linear embedding module projects the raw-valued features onto an arbitrary dimension. The number of embedding dimensions for the proposed model was set as 225 in this study.

Each SWINTF block comprises two successive SWINTF blocks, which consist of a window-based multihead self-attention (W-MSA) or shifted-window-based multihead self-attention (SW-MSA) partitioning module followed by two multilayer perceptron (MLP) modules, a layer norm (LN) module, and a residual connection. The first and second SWINTF block use a W-MSA and SW-MSA partitioning module, respectively. The MLP modules perform Gaussian error linear unit nonlinearity activation and contain 1024 neurons. The LN module is used before the MSA and MLP modules to normalize the distribution of the intermediate layers. The output features of the SWINTF block are calculated as follows [33]:

$$\begin{aligned} \hat{z}^{l} &= \mathrm{WMSA}\left(\mathrm{LN}\left(z^{l-1}\right)\right) + z^{l-1}, \\ z^{l} &= \mathrm{MLP}\left(\mathrm{LN}\left(\hat{z}^{l}\right)\right) + \hat{z}^{l}, \\ \hat{z}^{l+1} &= \mathrm{SWMSA}\left(\mathrm{LN}\left(z^{l}\right)\right) + z^{l}, \\ z^{l+1} &= \mathrm{MLP}\left(\mathrm{LN}\left(\hat{z}^{l+1}\right)\right) + \hat{z}^{l+1} \end{aligned} \tag{8}$$

where $\hat{z}^{l}$ and z^{l} denote the output features of the (S)W-MSA and MLP modules for block l, respectively and LN denote the layer normalization.

After the five SWINTF blocks have been used, the reshaping module is used. This module reshapes the feature maps output by the five SWINTF blocks by using a reshaping layer. Subsequently, it uses a depth-to-space layer and a pointwise 2-D convolutional layer to reconstruct the final output feature maps for the SWINTF submodel. Therefore, the output feature maps of the SWINTF submodel are 15 × 15 × 9 shape size.

The final layer in the proposed model is a linear pointwise 2-D convolutional layer that combines the feature maps output by the SWINTF submodel into a single canopy height per pixel. Overall, the proposed SST-CLT model has 11 216 557 trainable parameters, and the overall architecture of the proposed SST-CLT model is shown in Fig. 6. As recommended adaptive moment estimation with default parameter settings was used as the optimization function in the proposed model, and the learning rate was set as 0.0001 after the hyperparameters had been tuned [46]. The batch size was set as 32 during the training process, and the mean absolute error (MAE) was used as the loss function. The MAE is calculated as follows:

$$l = |y_i - \hat{y}_i| \tag{9}$$

where l, y_i, and $\hat{y}_i$ represent the loss function, actual value, and predicted value, respectively. The SST-CLT model was constructed using TensorFlow and Keras, which are provided by Google and was written in Python. The Python codes were executed on a Windows 10 platform with an Intel Core i9-11900K processor, an NVIDIA GeForce RTX 3090 graphics processing unit, and 64 GB of RAM.

D. Comparison and Evaluation

1) Comparison of the Proposed Model With Other Models: To determine whether the proposed model is effective, the prediction results obtained using the proposed model were compared with those obtained using other regression models, namely the multiple linear regression (MLr) model, decision tree regression (DTr) model, RF regression (RFr) model, Lang-CHM [9], Shah-CHM [36], and combination of vision-transformer with convolutional network (hereafter referred to as the C-VTC) inspired by Fayad et al. [47]. The Lang-CHM, Shah-CHM, and C-VTC are regression models based on deep learning architectures.

MLr is a traditional regression technique that is widely employed in many research fields. This method involves using two or more independent variables to predict a dependent variable by fitting a linear equation.

DTr is a supervised machine learning method that involves using decision rules and a tree-structured, decision-making process. A study adopted DTr to map the canopy height distribution in sub-Saharan Africa by using optical imagery [18]. In this study, we used the default function of the DTr algorithm from the scikit-learn package.

The RFr model is an ensemble tree-based classifier and uses major voting in the final classification to obtain the most popular class from the predicted results for each tree. Several studies have employed RF for vegetation height mapping [20], [21], [38]. In this study, we used 150 trees in our RFr algorithm.

The Lang-CHM is a supervised deep regression model that was originally developed by Lang et al. [9] for estimating country-wide (Switzerland and Gabon) forest canopy height by using Sentinel-2 imagery. The Lang-CHM comprises entry blocks and separable convolutional (SepConv) blocks. The entry blocks consist of three pointwise 2-D convolutional layers, whereas the SepConv blocks consist of a ReLU activation function layer followed by a depthwise separable convolutional layer and a batch normalization layer. In the Lang-CHM, data are passed through 18 SepConv layers before being input to a final pointwise convolution layer.

The Shah-CHM is a supervised deep-learning model adopted by Shah et al. [36] to estimate canopy height using Landsat imagery. This model contains four 2-D convolutional blocks. The first convolutional block consists of a convolutional layer followed by a dropout layer, a batch normalization layer, and a ReLU activation function layer. The second and third convolutional blocks consist of a convolutional layer followed by a batch normalization layer and a ReLU activation function layer. Finally, the last convolutional block comprises a convolutional layer followed by a batch normalization layer and a linear activation function layer.

The last baseline model comparison is a combined vision transformer with the convolutional network (C-VTC) inspired by Fayad et al. [47]. Similar to Lang et al. [9], the original model of Fayad et al. [47] was also used for forest canopy height estimation. We adjust the model [47] based on our input data for fair comparison, so that the input model has a shape width and height of 15 × 15. The C-VTC consists of an encoder-decoder block, the encoder block is transformer-based

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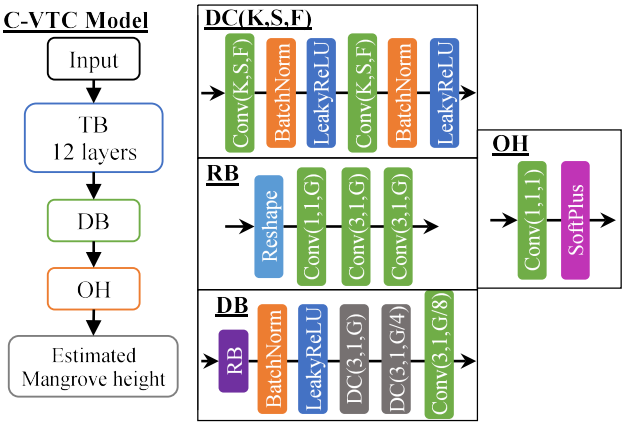


Fig. 7. Implementation of C-VTC with $G = 512$ inspired by Fayad et al. [47]. TB stands for vision-transformer block, DC (K, S, F) stands for deep convolutional (kernel, stride, and filters), RB stands for reshape block, DB stands for decoder block, and OH stands for output head.

architecture with 12 transformer layers while the decoder block is the convolutional network decoder (see Fig. 7). The transformer layer is a standard vision-transformer architecture [32] that consists of a normalization layer, followed by a multihead self-attention module, another normalization layer, and an MLP. For the decoder part, we adopted the convolutional decoder block 1 (DB1) from the [47] but we remove the two convolutional transpose layers because the implementation of C-VTC is without reducing the spatial dimensions during training, finally we set the initial filter size (G) of 512. For the output head, we adopted the single regression output [47] that consists of a single convolution operation with a kernel size of 1 and a stride of 1, followed by a Softplus activation layer. For the C-VTC model, we train the C-VTC model from scratch without a pretrained model. We use Huber loss as the loss function, Adam with a learning rate of $1e-5$ as the optimizer, and an early stop scheme using validation dataset.

To ensure a fair comparison, the same input data (Sentinel-1 and Sentinel-2 data) were used to train the five regression models; however, three schemes were applied. In the first scheme, Sentinel-1 and Sentinel-2 data (the original input data for this study) were employed; in the second scheme, the mean of Sentinel-1 and Sentinel-2 data was used; and in the third scheme, only Sentinel-2 data were employed. The performance comparison results are discussed in Section III.

2) Model Evaluation: After the training process, a novel testing dataset (20% of the total data) was used to evaluate the compared models regarding the MAE and root mean square error (RMSE). Sentinel-1 and Sentinel-2 data from the testing set were used as input data for the trained SST-CLT model to predict mangrove canopy height, and LiDAR CHM data were used as the ground-truth data. The MAE represents the average difference between the original and predicted values for the entire dataset, whereas the RMSE represents the standard deviation of the residuals (prediction errors). Lower MAE and RMSE values indicated better model performance. The MAE and RMSE were calculated as follows:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i| \tag{10}$$

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \hat{y}_i\right)^2} \tag{11}$$

where N is the number of data, y_i is the actual value, and $\hat{y}_i$ is the predicted value.

III. Results

A. Estimation of Mangrove Canopy Height by Using the SST-CLT Model

An independent testing dataset (20% of the total data) was used to evaluate the performance of the proposed model. The SST-CLT model was trained thrice, and the average MAE and RMSE were then calculated. For the ENP dataset (large dataset), the proposed model exhibited average MAE and RMSE values of 1.924 and 2.471 m, respectively. Moreover, for the CHPSP dataset (small dataset), the proposed model exhibited average MAE and RMSE values of 1.913 and 2.440 m, respectively. The average MAE was lower than 2 m for both datasets, which indicates that the SST-CLT model has a promising ability to estimate the height of a mangrove canopy when small or large datasets of Sentinel-1 time series and Sentinel-2 imagery are used.

The scatter plots of the ground-truth data (LiDAR CHM data) versus the prediction results for the ENP and CHPSP are displayed in Fig. 8. For the ENP, a strong agreement was discovered between almost all the predicted mangrove canopy heights and LiDAR CHM data. However, some canopy heights were overestimated when the vegetation height was low. For example, when the actual height was approximately 5 m, the predicted height exceeded 10 m. Moreover, underestimated results were obtained for actual heights exceeding 20 m. Such underestimation for high vegetation was also described in a previous study [9].

For the CHPSP, the predicted and actual canopy heights between 2 and 7 m were in close agreement. The predicted results for actual heights of less than 2 m tended to be overestimated, and many underestimated results were obtained for actual heights greater than 8 m. Thus, the SST-CLT model tended to misestimate the canopy height for actual heights greater than 8 m in the small dataset collected for the CHPSP. However, the MAE and RMSE values of this model for the CHPSP dataset were nonetheless acceptable.

To determine the effect of mangrove canopy height on the predicted results, the average MAE and RMSE were assessed for various canopy height ranges (see Tables IV and V). The testing data were divided using 5-m intervals. For the ENP dataset, the lowest average MAE (1.442 m) was obtained at a mangrove canopy height of 15–20 m, whereas the highest MAE (3.073 m) was obtained at a canopy height of >20 m. For the CHPSP dataset, only one pixel exhibited a mangrove height between 15 and 20 m, and no pixel exhibited a mangrove height of >20 m. The lowest average MAE (1.495 m) was obtained for a canopy height of 0–5 m, and the highest average MAE (5.251 m) was obtained for a canopy height of 10–15 m.

The results mentioned above were strongly affected by the proportions of the heights in the training sample, which contained fewer total training pixels for short and tall mangrove

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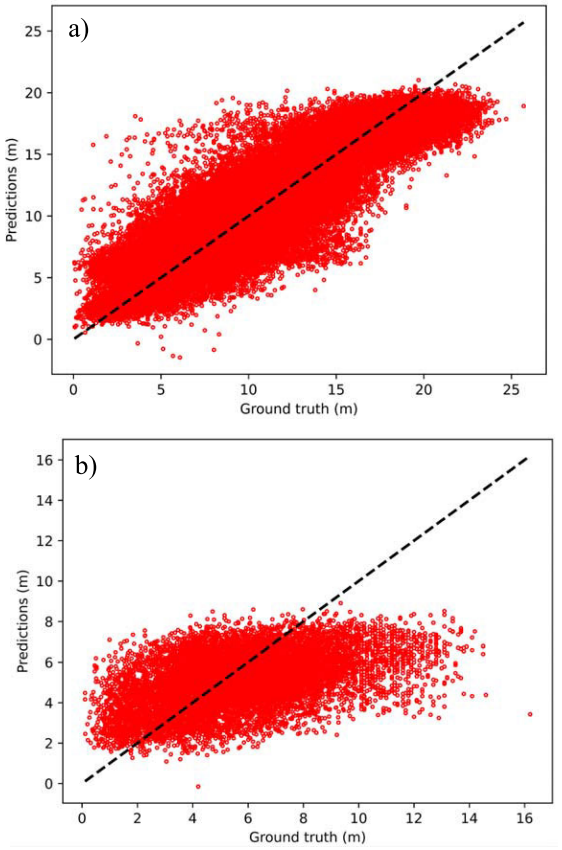


Fig. 8. Scatter plot of the ground-truth LiDAR CHM data (testing data) versus the prediction results: (a) ENP and (b) CHPSP.

TABLE IV
AVERAGE MAE VALUES OBTAINED WITH THE SST-CLT MODEL FOR VARIOUS HEIGHT RANGES IN THE ENP DATASET

Height (m)	Total Test Samples (pixels)	Avg. MAE (m)	Avg. RMSE (m)
0 – 5	5761	2.239	2.822
5 – 10	18779	1.696	2.235
10 – 15	17007	1.977	2.466
15 – 20	12444	1.442	1.888
> 20	2034	3.073	3.287

TABLE V
AVERAGE MAE VALUES OBTAINED WITH THE SST-CLT MODEL FOR VARIOUS HEIGHT RANGES IN THE CHPSP DATASET

Height (m)	Total Test Samples (pixels)	Avg. MAE (m)	Avg. RMSE (m)
0 – 5	4480	1.495	1.828
5 – 10	5663	1.738	2.134
10 – 15	656	5.251	5.438
15 – 20	1	12.774	12.774
> 20	0	0	0

vegetation than it did for average-height vegetation. Therefore, small and large mangrove heights were more difficult to predict than average mangrove heights. These results agree with those of a previous study [18] in which the effect of the high canopy reflectance of forest vegetation was reported to become saturated at a certain height and to then be constant.

B. Performance Comparison With Other Regression Models

The performance of the SST-CLT model was compared with that of other regression models (see Table VI). Performance was examined in terms of the average MAE and RMSE of the models applied to the testing dataset after three training sessions. The ranking of the compared models in terms of average MAE for the ENP and CHPSP datasets is shown in Fig. 9. For both datasets, the SST-CLT model outperformed the other regression models in terms of the average MAE and RMSE.

For the ENP dataset, the Lang-CHM with Sentinel-2 input data, the C-VTC with Sentinel-2 input data, the Shah-CHM with Sentinel-2 input data, the RFr model with the mean of the Sentinel-1 and Sentinel-2 data as input data, and the DTr model with Sentinel-2 input data obtained the second-best, third-best, fourth-best, fifth-best, and worst results, respectively. The MAE differences between these models and the proposed model were 0.29, 0.35, 0.44, 0.45, and 1.38 m, respectively. A reasonable explanation for these results is that the Lang-CHM, C-VTC, Shah-CHM, and proposed model consider spatial and spectral features, whereas the MLr, RFr, and DTr models only consider spectral features.

The CHPSP dataset is a small dataset that contains only 152 training images. For this dataset, machine-learning models, such as the RFr and MLr models, outperformed the deep-learning regression models, such as the Lang-CHM, C-VTC, and Shah-CHM. This finding indicates that deep-learning models that use only convolutional layers (e.g., the Lang-CHM and Shah-CHM) exhibit relatively poor performance for a small dataset. The C-VTC model that uses vision-transformer also has not too good performance in a small dataset but still has acceptable results referred to in the original paper [47] with MAE and RMSE of 2.462 and 3.219 m, respectively. However, the MAE of the Lang-CHM was still only 2.23 m, which is similar to the value obtained by Lang et al. [9]. By contrast, the proposed SST-CLT model uses spatial-spectral-temporal features extractor and SWINTFs that can extract local information and thus obtain promising results even for a small dataset. The proposed model exhibited the highest performance for the CHPSP dataset. The RFr model exhibited the second-best results, and the MAE difference between this model and the proposed model was only 0.033 m. The simple MLr model also achieved favorable results when the mean of the Sentinel-1 and Sentinel-2 data were employed as its input data (MAE = 2.053 m).

According to the experimental results, the Lang-CHM, C-VTC, and Shah-CHM were suitable for canopy height estimation when only Sentinel-2 data were used as the input data. This finding was obtained because, in the original study, only optical imagery (Sentinel-2 and Landsat data) was used as the input data [9], [36], [47]. When Sentinel-1 and Sentinel-2 data were used as the input data for the Lang-CHM, C-VTC, and Shah-CHM, the average MAE drastically increased; that is,

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TABLE VI
RMSE AND MAE VALUES OBTAINED WITH THE SST-CLT MODEL AND EXISTING MACHINE-LEARNING AND DEEP-LEARNING MODELS FOR THE ENP AND CHPSP DATASETS

Comparison with other regression model	ENP: Avg. MAE (m)	ENP: Avg. RMSE (m)	CHPSP: Avg. MAE (m)	CHPSP: Avg. RMSE (m)	Trainable Parameters
SST-CLT Model	**1.924**	**2.471**	**1.913**	**2.440**	**11,216,557**
MLr MeanS1+S2	2.778	3.522	2.053	2.604	-
MLr S1S2	2.778	3.522	2.054	2.604	-
MLr S2	2.898	3.660	2.063	2.617	-
DTr MeanS1+S2	3.217	4.218	2.533	3.201	-
DTr S1S2	3.249	4.258	2.548	3.237	-
DTr S2	3.310	4.352	2.604	3.277	-
RFr MeanS1+S2	2.375	3.100	1.946	2.476	-
RFr S1S2	2.384	3.105	1.953	2.483	-
RFr S2	2.463	3.208	1.961	2.496	-
Lang-CHM MeanS1+S2	2.648	3.366	2.244	2.857	19,618,273
Lang-CHM S1S2	3.113	3.916	2.278	2.909	19,618,785
Lang-CHM S2	2.222	2.883	2.230	2.843	19,618,017
Shah-CHM MeanS1+S2	2.752	3.472	4.865	5.500	41,177
Shah-CHM S1S2	2.603	3.324	5.391	6.001	50,393
Shah-CHM S2	2.365	3.023	4.314	4.956	36,569
C-VTC MeanS1+S2	2.427	3.138	3.022	3.856	22,566,829
C-VTC S1S2	2.800	3.575	3.105	3.930	22,589,329
C-VTC S2	2.276	2.962	2.462	3.219	22,555,579

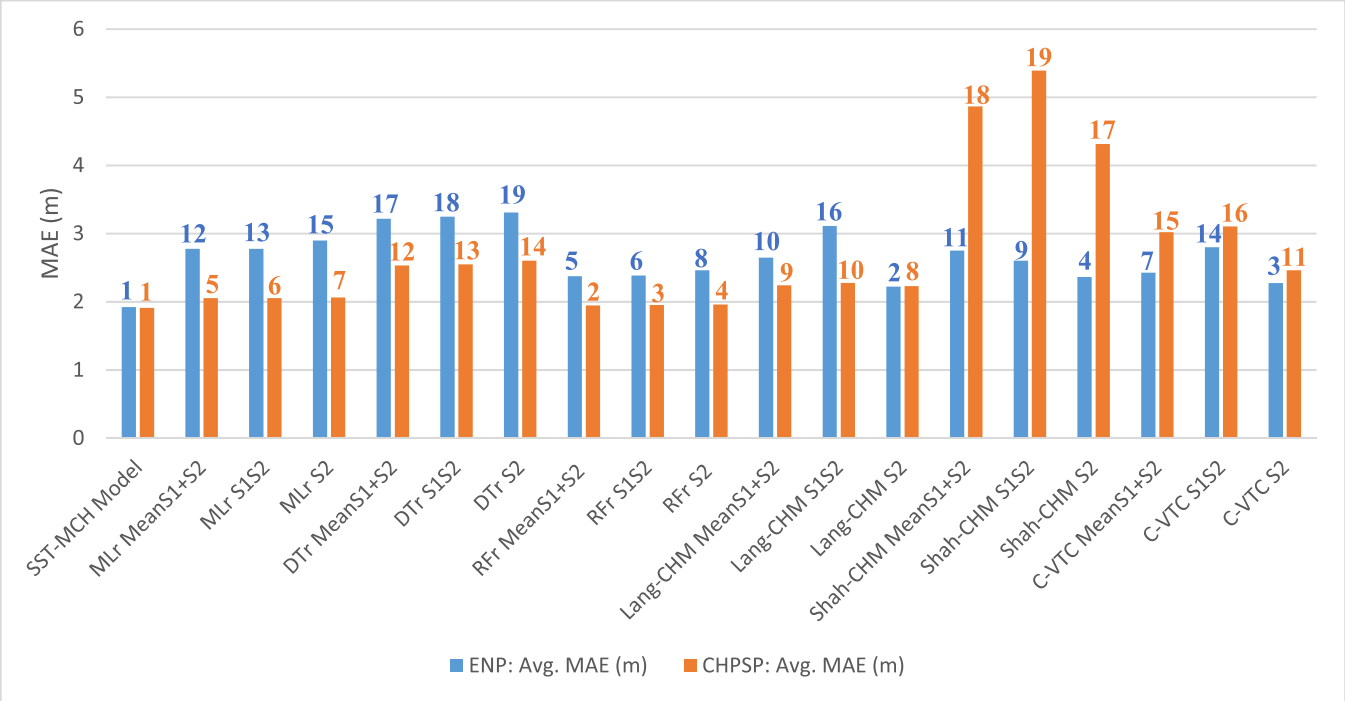


Fig. 9. Model ranking based on average MAE.

the models' performance drastically worsened. This occurred because the Lang-CHM, C-VTC, and Shah-CHM are not specifically designed to process the combination of Sentinel-1 and Sentinel-2 data. However, the SST-CLT model is specially

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TABLE VII
RESULTS OF THE ABLATION STUDY CONDUCTED ON THE SST-CLT MODEL

Without SST-S1 extractor	2.135	2.765
Replaced ConvLSTM2D with LSTM	2.070	2.677
Without SS-S2 extractor	3.484	4.483
Without SWINTF regressor sub-model	2.156	2.800
Ablation Study: Changed SWINTF regressor to SepConv or 2D Conv		
Changed with SepConv sub-model	2.019	2.612
Changed with Vision Transformer	2.059	2.666
Changed with 2D conv sub-model	2.038	2.631

designed to handle Sentinel-1 data (with the SST-S1 extractor) and Sentinel-2 data (with the SS-S2 extractor); thus, this model yielded more accurate canopy height estimations than did the other deep-learning regression models.

The SST-CLT model has 11 216 557 trainable parameters, which is fewer than the number for the Lang-CHM but considerably higher that the number for the Shah-CHM, which has very few trainable parameters because of its low complexity. However, according to our performance comparison results, the SST-CLT model considerably outperformed the other models. In particular, for the small CHPSP dataset, the SST-CLT model considerably outperformed the Lang-CHM, C-VTC, and Shah-CHM.

C. Ablation Study

To evaluate the effectiveness of the proposed model, an ablation study was performed using only the ENP dataset (because the CHPSP dataset was small). The ablation study aimed to determine the contributions of the two submodels to the proposed model's ability to predict mangrove canopy height (see Table VII). In total, we have seven ablation experiments: 1) removed the SST-S1 extractor part, 2) replaced the ConvLSTM2D layer with the LSTM layers in SST-S1 extractor part, 3) removed SS-S2 extractor part, 4) removed SWINTF submodel, 5) changed the SWINTF submodel with the SepConv submodel [9], 6) changed the SWINTF submodel with the 2-D convolutional block submodel [36], and 7) changed the SWINTF submodel with the original vision transformer (ViT) submodel [32].

When the proposed model was trained without the SST-S1 extractor, the average MAE and RMSE were 2.135 and 2.765 m, respectively. In advance, we replaced the ConvLSTM2D layers with LSTM layers in the SST-S1 extractor part to know the advantages of using ConvLSTM layers compared to LSTM layers. Based on the experiment result, using the ConvLSTM2D layers increased the performance of the proposed model. When we replaced the ConvLSTM2D layers with the LSTM layers, the average MAE and RMSE were increased by about 0.146 and 0.206 m, respectively.

The average RMSE obtained without the SST-S1 extractor was 0.294 m higher than that achieved with the complete SST-CLT model. The SST-CLT model was also trained without the SS-S2 extractor, which led to a considerable increase in the average MAE and thus a considerable decrease in performance. In addition, the SST-CLT model was trained without the SWINTF submodel, which resulted in a considerable decrease in the accuracy of the predictions. According to the experimental results, the RMSE increased by approximately 2.01 m when the SS-S2 extractor was removed, which indicates that this extractor is the most crucial part of the proposed model. The SST-S1 extractor and SWINTF submodel also improve the performance of the SST-CLT model.

The SWINTF submodel was replaced with three submodels used in previous studies: a SepConv submodel [9], a 2-D convolutional block submodel [36], and ViT blocks [32]. The results indicated that when the SepConv submodel was used instead of the SWINTF submodel, the average MAE and RMSE increased by 0.166 and 0.141 m, respectively. When the 2-D convolutional block submodel was used instead of the SWINTF submodel, the average MAE and RMSE increased by 0.144 and 0.160 m, respectively. Similarly, when the SWINTF blocks were replaced by the ViT blocks, the average MAE and RMSE increased by 0.135 and 0.195, respectively. Therefore, according to the experimental results, higher model performance was achieved when using the SWINTF submodel than when using the SepConv, ViT, or 2-D convolutional block submodel.

D. Effects of Various Input Bands

The final input bands for the SST-CLT model were Sentinel-1 time-series data (VV and VH data) and Sentinel-2 data (the blue, green, red, NIR, SWIR-1, SWIR-2, and NDVI bands). Seven experiments were conducted to determine the most favorable input data combination and the effects of using Sentinel-1 time series and Sentinel-2 bands in the proposed model (see Table VIII). The best model is the first experiment; the original input data were used to confirm whether the Sentinel-1 and Sentinel-2 data improved the predicted results and yielded the most favorable outcomes. In the second experiment, the NDVI band was removed from the input data, and the results indicated that this resulted in increases in the average MAE of only 0.079 and 0.024 m for the ENP and CHPSP datasets, respectively. Thus, the second-best model performance was achieved with the aforementioned data. The NDVI band was also removed in the third and fourth experiments. In these experiments, attempts were made to determine the effect of using SWIR-1 or SWIR-2 alone. The results indicated that more accurate predictions were obtained for the ENP and CHPSP datasets when using SWIR-1 than when using SWIR-2 ENP.

In the fifth experiment, the SWIR-1 and SWIR-2 bands were removed from the input data, and the results indicated that the use of these bands improved the predicted outcomes. In the sixth experiment, the NDVI and SWIR bands were removed, and this resulted in the average MAE increasing by 0.166 and 0.331 m for the ENP and CHPSP datasets, respectively. The final experiment was conducted using only Sentinel-1 time-series data with RGB images (i.e., removing the NIR, SWIR, and NDVI bands from the input data). The average MAE and RMSE increased in this experiment; thus, the original input

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TABLE VIII
EFFECTS OF VARIOUS INPUT BANDS ON THE PREDICTION PERFORMANCE OF THE SST-CLT MODEL FOR THE ENP AND CHPSP DATASETS, RESPECTIVELY (B, G, R, AND N REPRESENT THE BLUE, GREEN, RED, AND NIR BANDS, RESPECTIVELY)

Input Bands	ENP MAE (m)	CHPSP MAE (m)
SST-CLT model: S1 + S2 (B-G-R-N-SWIR1-SWIR2-NDVI)	**1.924**	**1.913**
S1+B-G-R-N-SWIR1-SWIR2	2.003	1.954
S1+B-G-R-N-SWIR1	2.016	2.038
S1+B-G-R-N-SWIR2	2.027	2.102
S1+B-G-R-N-NDVI	2.049	2.120
S1+B-G-R-N	2.090	2.229
S1+B-G-R	2.234	2.244

bands improved the predictions of the proposed model for the ENP and CHPSP datasets.

IV. DISCUSSION

A. Map of Mangrove Canopy Height for the ENP

The trained SST-CLT model was used to determine the mangrove canopy height in the ENP. All Sentinel-1 and Sentinel-2 images covering the entire study area were used as input data for this model. To obtain large-scale maps, Sentinel-1 and Sentinel-2 images were first cut into image tiles of size 15 × 15 pixels, and a three-pixel overlap was used to overcome the tiling artifact problem. Next, the 15 × 15-pixel image tiles covering the entire mangrove area of the ENP were employed in the proposed model for predicting the mangrove canopy height. Finally, these tiles were recomposed into a large image to obtain the mangrove canopy height in the entire area of the ENP.

The map of mangrove canopy height for the ENP in 2017 had a spatial resolution of 10 m (see Fig. 10). According to this map, the average mangrove canopy height in the ENP was 7.68 m in 2017. A short mangrove area was found bordering the land. Most of the mangrove area had a canopy height of approximately 8–11 m and was located bordering the coastal area. Most of the mangroves in the northwestern area had a canopy height of approximately 8–14 m, whereas most of the mangroves in the southeastern area had a canopy height of approximately 1–10 m. In Fig. 10, the area with the tallest mangroves is indicated by a red box, which represents the Shark River Slough (SRS) area. In this area, the mangrove canopy height mostly exceeded 14 m. As mentioned by Feliciano et al. [5] and Simard et al. [12], the SRS area had a taller mangrove canopy than the other areas in the ENP.

The mangrove canopy height obtained in this study for the ENP was compared with the findings of two previous studies (see Fig. 11). The first study successfully obtained the mangrove canopy height in the ENP during 2000–2004 at a spatial resolution of 30 m, whereas the second study obtained the mangrove canopy height in the ENP during 2011–2014 at a spatial resolution of 12 m.

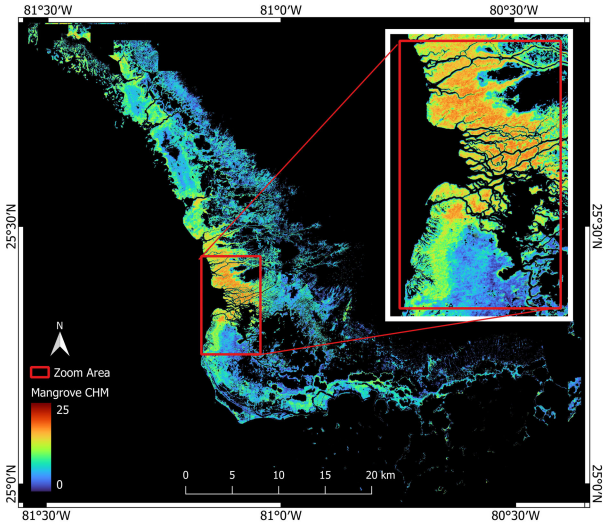


Fig. 10. Map of mangrove canopy height in the ENP in 2017 at a spatial resolution of 10 m.

A visual comparison of the three maps indicated that the three maps exhibited similar patterns in mangrove height. The mangrove canopy height in the SRS region (zoomed-in view area) increased from 2000 to 2017. In 2000, this region mostly had a mangrove canopy height of approximately 14–16 m. In 2014, the mangrove canopy height was still mainly 14–16 m; however, many mangroves taller than 18 m were detected. In addition, the number of mangroves taller than 18 m increased further between 2014 and 2017.

B. Mangrove Canopy Height Map for the CHPSP

The trained SST-CLT model was used to produce a mangrove canopy height map for the CHPSP (see Fig. 12). The steps involved in producing this map were the same as those involved in producing the mangrove canopy height map for the ENP. The final mangrove canopy height map for the CHPSP (for 2017) had a spatial resolution of 10 m. According to this map, the average mangrove height in the CHPSP was 4.381 m. Almost all the mangroves in this study area had a canopy height of 2–6 m.

We compared the aforementioned mangrove map with the global mangrove canopy height obtained for the CHPSP from the dataset of the NASA Carbon Monitoring System (CMS) [12]. This dataset has a spatial resolution of 30 m and temporal coverage of 2000–2009. The average mangrove canopy height determined using the CMS dataset was 7.54 m. The dominant mangrove canopy height obtained from this dataset was approximately 2–8 m.

The dominant mangrove canopy height obtained from the CMS dataset differed from that in our mangrove canopy height map for the CHPSP. However, as presented in Table III, the average mangrove canopy height in the CHPSP according to the mangrove CHM LiDAR data was 4.85 m, which is similar to the mangrove canopy height in our map for the CHPSP in 2017 (4.381 m).

C. SST-CLT Model Performance Analysis

The proposed SST-CLT model has the ability to consider the temporal information from satellite imagery, as mentioned

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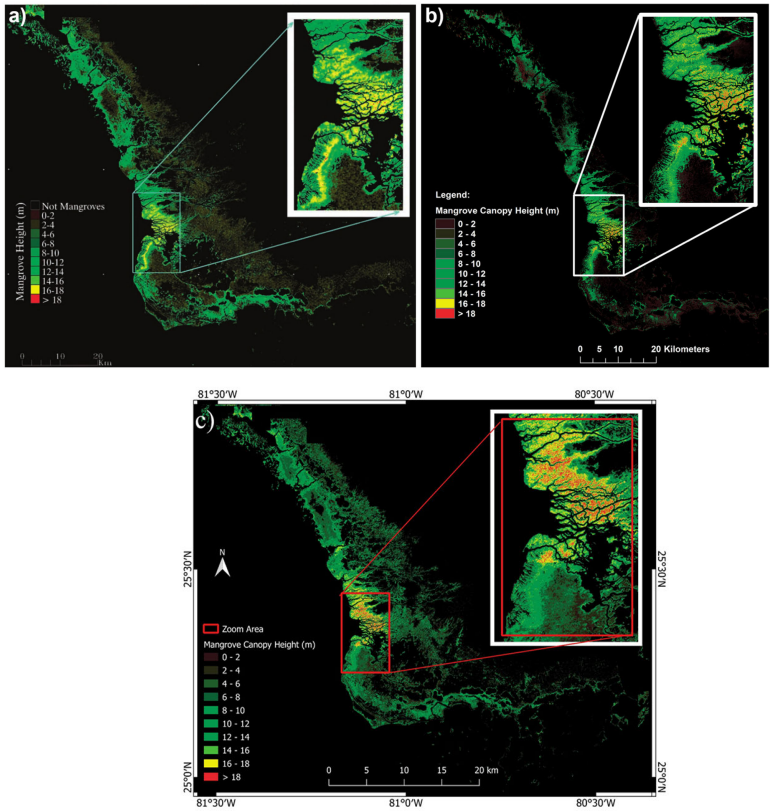


Fig. 11. Maps of mangrove canopy heights in the ENP at different times: (a) during 2000–2004 at a spatial resolution of 30 m (published by Simard et al. [12]), (b) during 2011–2014 at a spatial resolution of 12 m (published by Feliciano et al. [5]), and (c) in 2017 at a spatial resolution of 10 m (obtained using the proposed model).

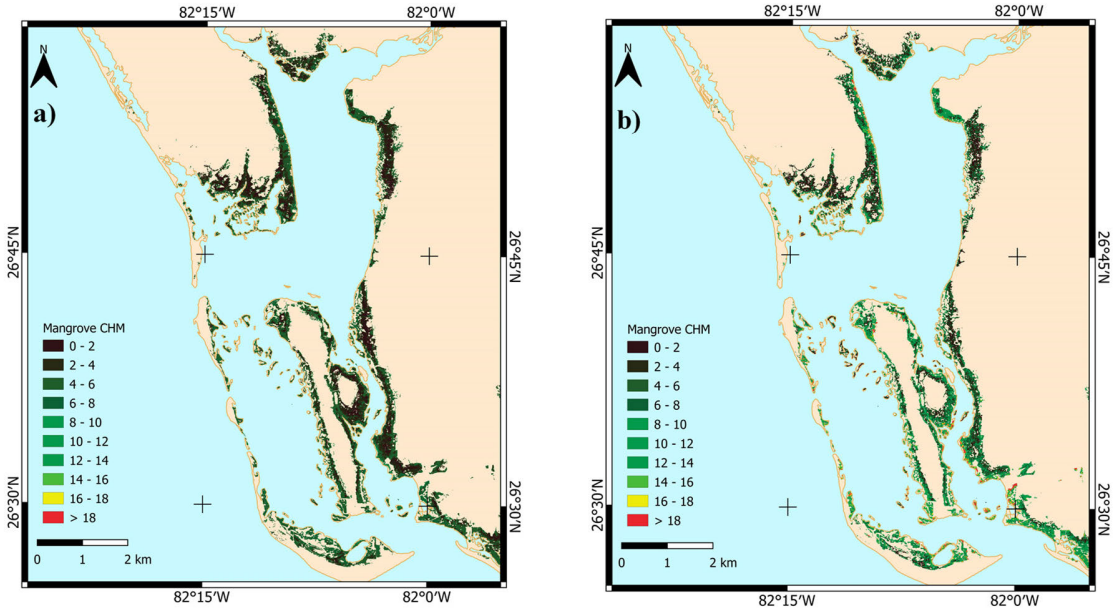


Fig. 12. Comparison of the mangrove canopy heights obtained for the CHPSP in this study and a previous study: (a) map for 2017 with a spatial resolution of 10 m that was obtained in this study and (b) map for 2000–2009 with a spatial resolution of 30 m that was obtained by Simard et al. [12].

in the previous study, that spectral time series could correlate even better with the vegetation canopy height [9]. The experiment result shows our SST-CLT model that considers the spatial-spectral-temporal information from satellite imagery has a better performance than the other regression model that just considers spatial-spectral information (Lang-CHM, C-VTC, and Shah-CHM) or just spectral information (RFr, DTr, and MLr).

The use of the SWINTF submodel in our SST-CLT model increased the model's performance because the SWINTF can

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TABLE IX
COMPARISON OF MODEL'S EFFECTIVENESS BASED ON TRAINING TIME AND SECONDS PER-EPOCH (SPE) USING ENP DATASET

Model	Best MAE (m)	Training Time (mm:ss)	SPE (s)
SST-CLT	1.924	05:29	3.59
Lang-CHM	2.222	04:55	2.77
Shah-CHM	2.365	00:49	0.17
C-VTC	2.276	09:59	2.91
RFr	2.375	05:56	-
DTr	3.217	00:02	-
MLr	2.778	00:01	-

also consider the local pixel value of mangrove canopy height, which is very suitable for our pixel regression task. This was proven by comparing the SWINTF submodel with the ViT submodel in the ablation study part. The SWINTF block outperformed the ViT block because the hierarchical partitioning helps to capture both local and global information effectively.

Overall, the proposed SST-CLT model demonstrates higher performance in terms of MAE and RMSE evaluation than the state-of-the-art regression models. However, the execution training time and second per-epoch (SEP) time also need to be investigated to know the effectiveness of the proposed model. The comparison of the model's effectiveness based on the average total training time and SPE using the ENP dataset is shown in Table IX. In general, our proposed model has acceptable training time with a total average training time of about 5 min (05:29—mm:ss) with an SPE of 3.59 s. The average total training time of SST-CLT is faster than the RFr model. However, there is a slight difference between the training time for SST-CLT and C-VTC model. While the simpler model, such as Shah-CHM, DTr, and MLr, has fast training time but the performance is lower than the SST-CLT model. The SWINTF has self-attention mechanisms that can cause an increase in model complexity. The C-VTC model that also use transformer layer and has SPE 2.91 s which is higher than Lang-CHM but lower than our proposed model. Nevertheless, our SST-CLT model achieved more accurate results that can answer our main goal of producing accurate large-scale mangrove canopy height and also has reasonable training and inference time.

The trained SST-CLT model can produce a large-scale mangrove canopy height at 10-m spatial resolution with more accurate results (MAE below 2 m). We have proven that the trained model can produce large-scale mangrove canopy height in ENP and CHPSP areas. This study opens up the possibility for global 10-m spatial resolution mangrove canopy height mapping. The main challenge is how to collect the LiDAR CHM of mangroves in other locations. The spaceborne LiDAR data [e.g., Global Ecosystem Dynamics Investigation (GEDI)] may also be used for estimating global mangrove canopy height in future studies. On the other hand, we acknowledge the potential benefits of leveraging the red-edge bands from Sentinel-2 for obtaining vegetation information, especially when the NDVI value is saturated with high vegetation density. The comprehensive experiments and analysis incorporating the red-edge bands as the additional input bands and for NDVI calculation in our model will be considered for future work.

V. CONCLUSION

This study developed a deep-learning regression architecture called the SST-CLT model for the large-area mapping of mangrove canopy height by using Sentinel-1 time-series data and Sentinel-2 data. The SST-CLT model consists of two submodels: the feature extractor submodel with two parts (the SST-S1 and SS-S2 extractors) and the SWINTF submodel.

The SST-CLT model was tested on two datasets: a large dataset (for the ENP) and a small dataset (for the CHPSP). The results indicated that the SST-CLT model successfully combined Sentinel-1 time-series data with Sentinel-2 data as input data for both datasets. The testing results revealed that the average MAE of the mangrove canopy heights obtained using the proposed model was 1.924 and 1.913 m for the ENP and CHPSP datasets, respectively.

The performance of the proposed model was compared with that of other regression models, and the results indicated that the proposed model yielded more accurate results in terms of the average MAE and RMSE than the other regression models. An ablation study was performed to determine the contributions of different parts of the model to its performance. The SS-S2 extractor was determined to be the most crucial part of the proposed model. Nevertheless, the SST-S1 extractor and SWINTF submodel also contributed to the high performance of the SST-CLT model.

We examined the influences of various input bands on the performance of the SST-CLT model. The most favorable input data for the SST-CLT model were Sentinel-1 time-series data (VV and VH data) and Sentinel-2 data (the blue, green, red, NIR, SWIR-1, SWIR-2, and NDVI bands). When any one of the input bands was removed, the performance of the proposed model decreased.

Finally, this study investigated the mangrove canopy height in the entire areas of the ENP and CHPSP in 2017 at a spatial resolution of 10 m. According to the obtained mangrove canopy height maps, the average mangrove canopy heights in the ENP and CHPSP were 7.68 and 4.381 m, respectively. In the ENP, most of the tall mangroves were located in the SRS area, whereas the areas with the shortest mangroves mostly bordered the land. The mangrove canopy height map obtained for the ENP in this study was compared with those reported in two previous studies, in which the investigation periods were 2000–2004 and 2011–2014, respectively. All three maps exhibited similar patterns in mangrove canopy height, and the mangrove height was discovered to have increased from 2000 to 2017. Moreover, the mangrove canopy height map obtained for the CHPSP in this study was compared with the mangrove canopy height data obtained from the CMS dataset. The average mangrove canopy height differed between the map obtained in this study and the CMS dataset.

The SST-CLT model achieved low MAE and RMSE values in the mapping of mangrove canopy height when using large

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and small datasets. However, this model was trained using airborne LiDAR CHM data, which have limited coverage. A future study may employ spaceborne LiDAR data to obtain greater global coverage.

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